Macroeconomics I — MPE Insper

Practice list: Every topic of the course (Aulas 1–9)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

Block 1 — Measuring aggregates: GDP, price indices, PPP, beyond GDP

(Aula 1 • Kurlat chs. 1–2 • 10 points)

Question 1. (4 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 1.3 "Chained GDP", pp. 28–29 — statement reproduced verbatim, all items (a)–(c).*

Context from the book (added by this list, from Kurlat §1.2, "Making Comparisons", pp. 23–25): *Base-year prices* (Alternative 1, p. 24): real GDP in year t is

$$Y_t = \sum_i p_{i0}\, q_{it}, \tag{1.2.1}$$

where Y_t is real GDP in year t, p_{i0} is the price of good i in the base year (year 0) and q_{it} is the quantity of good i produced in year t. *Chained prices* (Alternative 3, pp. 24–25): start from base year 0; compute real growth between consecutive years in two ways, at the earlier year's and at the later year's prices; average the two growth rates; add the average growth to the previous year's real GDP; repeat for every year. The book's formulas (unnumbered, p. 25):

$$g_t^I = \frac{\sum_i p_{it-1}\, q_{it}}{\sum_i p_{it-1}\, q_{it-1}} - 1, \qquad g_t^F = \frac{\sum_i p_{it}\, q_{it}}{\sum_i p_{it}\, q_{it-1}} - 1,$$

$$g_t = \left(1 + g_t^I\right)^{0.5} \left(1 + g_t^F\right)^{0.5} - 1, \qquad Y_t = Y_{t-1}\,(1 + g_t),$$

with g_t^I growth based on initial-year prices, g_t^F growth based on final-year prices, g_t their geometric average and Y_t real GDP one year ahead; the result is GDP in "chained" prices of the base year.

The country of Fructus produces Apples, Bananas and Cherries. Its production statistics are given below:

	Apples		**Bananas**		**Cherries**	
	quantity	price	quantity	price	quantity	price
2000	100	40	50	30	200	20
2001	103	39	47	35	206	20
2002	107	39	47	38	213	19
2003	109	39	45	38	215	18
2004	112	40	46	41	224	18
2005	112	39	46	39	227	17
2006	114	40	47	36	237	16
2007	115	40	46	42	249	16
2008	119	40	43	44	255	16
2009	118	42	40	46	262	15
2010	116	43	41	44	268	15
2011	118	42	40	50	280	15
2012	117	42	41	50	289	15
2013	119	43	40	53	290	14
2014	120	44	40	56	297	13
2015	125	45	41	59	308	13
2016	128	45	38	66	320	13
2017	129	46	38	61	330	13
2018	131	47	37	64	344	13
2019	136	47	37	61	353	13
2020	143	47	34	61	361	12

(you can download them as an Excel spreadsheet from the book website)

(a) Compute a real GDP series at year-2000 prices using base year prices and using the chained method.

(b) Plot both series and comment on any differences.

(c) What was the average growth rate according to each method?

Added to the book's items (not in the book): use the chained method of Kurlat §1.2; show the 2000–2001 link in full, describe the plot of (b) in words, and report both series for 2000, 2005, 2010, 2015 and 2020 and the average growth rates.

Solution:

Question 2. (5 points) Discursive. *Three approaches × price indices × beyond GDP.* An illustrative Brazilian economy (BRL billion, 2025) has four producers.

- Agro firm F grows soybeans with no intermediate inputs. It sells 300 to the food processor P, exports 200 and pays wages of 250.
- P buys the 300 of soybeans, imports packaging for 100 and buys 40 of online advertising from N. It produces 700 of food: households buy 650 and 50 stays unsold in P's warehouse. P pays wages of 180.
- Platform N gives households a free app financed by advertising; its only sale is the 40 of ads to P, and it pays 30 to the app staff. It also pays 60 in wages to programmers who write software that N will use in its own operations for ten years.
- The government pays 120 in wages to public school teachers and 80 in Bolsa Família transfers, financed by taxes.

Each firm's profit is its revenue plus any output added to its own stocks, minus wages and intermediate inputs.

a) (2 points) Compute 2025 GDP by the production, income and expenditure approaches, and show that they agree.

Solution:

b) (1.5 points) Give the number and the reason in each case. (i) GDP if N books the software as

a current expense. (ii) GDP a student obtains by counting Bolsa Família in G. (iii) Gross national income if 30 of F's profit belongs to a foreign shareholder. (iv) What part of the app's value to households GDP records, and how a welfare measure such as Jones–Klenow's would treat it.

Solution:

c) (1.5 points) In 2026 every quantity repeats, while prices change: food +10%, soybeans +20% (domestic and export price), imported packaging +50%, teachers' wages +5%; ads and software unchanged. Compute nominal GDP in 2026, real GDP at 2025 prices and the inflation rate of the GDP deflator. Compare it with a consumer price index (IPCA-type, Laspeyres) built on the household basket, and explain the gap.

Solution:

Question 3. (1 point) Multiple choice. Brazil's GDP per capita is BRL 60,000 (illustrative); the market exchange rate is 5.0 BRL per USD and the Penn World Table PPP conversion factor is 2.5 BRL per USD; US GDP per capita is USD 85,000. Which statement is correct?

A. At PPP Brazil's GDP per capita is USD 12,000, about 14% of the US level, since PPP strips out the cheaper Brazilian tradables.

B. At PPP it is USD 24,000, about 28% of the US level, because tradable goods are cheaper in Brazil while the real is undervalued.

C. At PPP it is USD 24,000, about 28% of the US level, because non-tradable services are cheaper in Brazil, so a real buys more there.

D. At PPP it is USD 150,000, about 176% of the US level, because PPP converts reais at the price of a common basket of goods.

E. At market rates it is USD 24,000 and at PPP USD 12,000, so the market rate overstates Brazilian living standards.

Block 2 — The Solow model: mechanics

(Aula 2 • Kurlat ch. 3, §4.1–4.2 • 10 points)

Question 4. (4 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 4.2 "The Black Death", pp. 72–73 — statement reproduced verbatim, all items (a)–(d).*

Context from the book (added by this list, from Kurlat §4.1, §4.2 and §4.4, pp. 53–69): *The Solow model.* Output is $Y = F(K, L)$ (4.1.1), with K capital and L labor; F has constant returns to scale and positive, diminishing marginal products ($F_K, F_L > 0$; $F_{KK}, F_{LL} < 0$); the Cobb–Douglas case (p. 55) is

$$Y = K^{\alpha} L^{1-\alpha}. \tag{4.1.2}$$

Population grows at the exogenous rate n, $L_{t+1} = (1+n)L_t$, and everyone alive works, so L is both the population and the labor force (Assumption 4.5, p. 55). The economy is closed with no government, $Y = C + I$, and saves a constant fraction s of output, $I = sY$ (4.1.3, p. 56). Capital depreciates at rate δ (p. 57):

$$K_{t+1} = (1-\delta)\, K_t + I_t, \tag{4.1.4}$$

so each period's capital stock is inherited from the past. Per worker, $y \equiv Y/L$, $k \equiv K/L$, $y = f(k) \equiv F(k, 1)$ (4.2.1), and (p. 58)

$$k_{t+1} - k_t = \frac{s f(k_t) - (\delta + n)\, k_t}{1+n}. \tag{4.2.2}$$

Prices (§4.4, pp. 66–69): competitive firms pay the rental rate of capital $r^K = F_K(K, L)$ (4.4.8) and the wage $w = F_L(K, L)$ (4.4.9); lending and investing must pay the same, so the real interest rate on a loan from t to $t+1$ is

$$r_{t+1} = r^K_{t+1} - \delta. \tag{4.4.12}$$

In the middle of the 14th century, a plague killed about a third of the population of Europe. Assume that the economy of Europe was well described by the Solow model. How would the following variables change in response to the Black Death in the short run (i.e. immediately)?

(a) GDP,

(b) GDP per capita,

(c) wages,

(d) interest rates.

Added parts (not in the book): use $Y = AK^{\alpha}L^{1-\alpha}$; give each change as a factor and, with $\alpha = 1/3$, as a percentage. Then add one line per variable on the long run, with s, n, δ and A unchanged.

Solution:

Question 5. (5 points) Discursive. *Steady state × saving reform × convergence.* Per-worker variables throughout ($k = K/L$, $y = Y/L$, $c = C/L$). $Y_t = AK_t^{\alpha}L_t^{1-\alpha}$, $K_{t+1} = (1-\delta)K_t + sY_t$, $L_{t+1} = (1+n)L_t$, no technological progress. Illustrative calibration for Brazil: $A = 1$, $\alpha = 1/3$, $s = 0.20$, $n = 0.02$, $\delta = 0.08$.

a) (2 points) Derive the law of motion of k_t and find k^*, y^* and c^* in closed form in $(A, s, n, \delta, \alpha)$, then in numbers. Use the limit $\alpha \to 0$ to check the exponent of y^*.

Solution:

b) (1.5 points) A pension reform raises s to 0.25 for good. Find the new y^* and its percentage change, the growth of k and y in the first period after the reform, consumption on impact and in the new steady state, and the long-run growth rate of y. Draw the Solow diagram.

Solution:

c) (1.5 points) Two states share the calibration of (a): São Paulo sits at k^* and Piauí starts at $k^*/8$. Find each state's growth of k and y in the first period. Then let Piauí save only $s = 0.10$: find its own steady state and its first-period growth, and say what happens to y_{PI}/y_{SP}. Which kind of convergence does each case illustrate?

Solution:

Question 6. (1 point) Multiple choice. Brazil's demographic transition lowers population growth n from 2% to 1% a year for good. In the Solow model without technological progress, starting and ending in steady state:

A. k^* and y^* rise; long-run growth of y stays at zero, while total GDP growth falls from 2% to 1% a year.

B. k^* and y^* rise; long-run growth of y rises by one point, while total GDP growth stays at 2% a year.

C. k^* and y^* fall; long-run growth of y stays at zero, while total GDP growth stays at 2% a year.

D. k^* and y^* rise; long-run growth of y stays at zero, while total GDP growth stays at 2% a year.

E. k^* rises and y^* stays put; long-run growth of y stays at zero, while total GDP growth falls from 2% to 1% a year.

Block 3 — Solow continued: golden rule, TFP and growth accounting

(Aula 3 • Kurlat §4.3–4.5, ch. 5 • 10 points)

Question 7. (4 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 5.4 "Measurement of the Capital Stock", pp. 94–95 — statement reproduced verbatim, all items (a)–(d).*

The capital stock is not easy to measure. A common way to try to do this is to:

- guess a value for K_0 at some date in the past (maybe when statistics were first collected),
- measure investment every year thereafter (which is also not that easy), and
- use the equation

$$K_{t+1} = (1 - \delta) K_t + I_t \tag{5.5.1}$$

to compute the capital stock for every year thereafter.

Consider an economy that is in a steady state without technological progress (and has been there for an infinitely long time). Every year the level of investment is:

$$I_t = 0.2$$

and the depreciation rate is:

$$\delta = 0.1$$

(a) What is the capital stock?

(b) The statistical office only started measuring investment in year 0. By then the economy was already in steady state. They guessed (incorrectly) that the capital stock in year 0 was $K_0 = 1$. Use equation (5.5.1) to compute how the estimate of the capital stock changed over time after year 0 and plot your answer. (You don't need to provide a closed form solution, a nice graph made with a spreadsheet is enough). Compute the ratio:

$$\frac{K_{ESTIMATE}}{K_{TRUE}}$$

for the years 5, 10, and 50. (Again, you can just take the numbers from a spreadsheet).

(c) Suppose now that the statistics office started collecting statistics a long time ago (an infinite time ago) so we don't have the problem of making the wrong initial guess. However, instead of using the correct value of δ, the statistical office incorrectly believes that $\delta = 0.05$. What is their estimate of the capital stock? Why does it differ from the truth?

(d) The production function in this economy is

$$Y = AK^{\alpha}L^{1-\alpha} \tag{5.5.2}$$

and there are accurate data on GDP and on total hours of labor supply. Using the inaccurate estimate of K from part (c), plus accurate data on total labor and GDP, an economist is trying to measure the economy's productivity, i.e. to solve for A in equation (5.5.2). Find an expression for

$$\frac{A_{ESTIMATE}}{A_{TRUE}}$$

as a function of

$$\frac{K_{ESTIMATE}}{K_{TRUE}}$$

Given the numbers from part (c) and using $\alpha = 0.35$, how far would the economist's estimate be from the truth?

Changed by this list (not in the book): for (b), give the closed form of the estimate, not only a spreadsheet.

Solution:

Question 8. (5 points) Discursive. *Golden rule* × *interest rates* × *dynamic efficiency.* Solow model in continuous time with labour-augmenting technology: $Y = K^{\alpha}(AL)^{1-\alpha}$, $\dot{A}/A = g$, $\dot{L}/L = n$, depreciation δ, saving rate s. Variables per efficiency unit: $\tilde{k} = K/(AL)$, $\tilde{y} = \tilde{k}^{\alpha}$, $\tilde{c} = (1-s)\tilde{y}$, and $\dot{\tilde{k}} = s\tilde{k}^{\alpha} - (n+g+\delta)\tilde{k}$. Competitive factor markets give $r = \alpha\tilde{k}^{\alpha-1} - \delta$. Illustrative calibration for Brazil: $\alpha = 1/3$, $n = 0.01$, $g = 0.02$, $\delta = 0.05$, $s = 0.18$.

a) (2 points) Find steady-state $\tilde{c}^*(s)$ in closed form and the saving rate s_{GR} that maximises it, with the second-order condition. Cross-check with the capital-stock version of the problem, and use a knife-edge value of α to test your exponent.

Solution:

b) (1.5 points) Compute $\tilde{k}^*$, $\tilde{c}^*$ and r at $s = 0.18$ and at s_{GR}. Show that $r - (n+g)$ has the sign of $\alpha - s$, and say whether Brazil should raise its saving rate.

Solution:

c) (1.5 points) Another economy has the same parameters but $s = 0.45$. Show that cutting s to α raises consumption per efficiency unit on impact and at every later date, and name the inefficiency.

Solution:

Question 9. (1 point) Multiple choice. Illustrative Brazilian data, 2000–2014 averages: GDP grows 3.2%, the capital stock 4.4% and hours 1.8% a year; the capital share is 0.4. The Solow residual, and the share of the growth of output *per hour* it accounts for, are:

A. 0.36 p.p. a year, accounting for about 11% of the growth of output per hour.

B. −0.16 p.p. a year, accounting for about −5% of the growth of output per hour.

C. 1.44 p.p. a year, accounting for about 45% of the growth of output per hour.

D. 0.36 p.p. a year, accounting for about 26% of the growth of output per hour.

E. 0.56 p.p. a year, accounting for about 40% of the growth of output per hour.

Block 4 — Consumption and saving

(Aula 4 • Kurlat ch. 6 • 10 points)

Question 10. (4 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 6.4 "Consumption and Income", pp. 122–123 — statement reproduced verbatim, all items (a)–(d).*

Context from the book (added by this list, from Kurlat §6.1, pp. 103–104, and Exercise 6.1, p. 121): *Exercise 6.1* ("Two-Period Problem with Taxes and Initial Wealth"): a household solves

$$\max_{c_1,a,c_2} \; u(c_1)+\beta u(c_2) \quad \text{s.t.} \quad a = a_0 + y_1 - \tau_1 - c_1, \qquad c_2 = y_2 - \tau_2 + (1+r)\,a,$$

with $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, where c_1, c_2 are consumption at times 1 and 2, y_1, y_2 household income at times 1 and 2, τ_1, τ_2 taxes at times 1 and 2 and a_0 initial wealth; a is the wealth carried from time 1 to time 2, r the interest rate and β the discount factor. *The Keynesian view* (§6.1): consumption C is a function of income Y,

$$C = c(Y), \tag{6.1.1}$$

with a marginal propensity to consume $c'(Y) < 1$ (when income rises by one dollar, consumption rises but by less than one dollar) and, on a stronger reading, an elasticity $\frac{c'(Y)\,Y}{c(Y)} < 1$. The book tests it by fitting (6.1.1) to a cross-section of households' income and consumption.

Suppose there are two households in the economy. Each of them solves a special case of Exercise 6.1, where $\tau_1 = \tau_2 = a_0 = 0$, $\beta = 1$ and $r = 0$.

(a) Solve for c_1 as a function of y_1 and y_2.

(b) Suppose the incomes of each of the households are:

	y_1	y_2
Household A	2	4
Household B	6	4

Compute c_1 and c_2 for each of the households.

(c) Suppose an economist is trying to decide what is a reasonable model for consumption behavior and only has data for period 1. Is the data supportive of the Keynesian view of the consumption function? Explain.

(d) Suppose the same economist now looks at data for period 2 in addition to data for period 1. Is the data supportive of the Keynesian view of the consumption function? Explain.

Added to the book's items (not in the book): in (c) and (d), fit the line $c = a + by$ that a Keynesian economist would estimate and say what it does and does not show.

Solution:

Question 11. (5 points) Discursive. *Choice variables* × *Ricardian equivalence* × *credit constraints.* A Brazilian worker lives two periods (working life and retirement, about 30 years each) with utility $\ln c_1 + \beta \ln c_2$, $\beta = 0.6$. In period 1 she holds initial wealth $a_0 = 10$, earns labour income $w_1 = 40$ (hours are fixed), receives dividends $\Pi_1 = 10$ from the family firm she owns and pays a lump-sum tax $\tau_1 = 10$. In period 2 she receives income $y_2 = 90$ and pays $\tau_2 = 10$. She borrows or lends a at the gross rate $1 + r = 2$ (about 2.3% a year over 30 years).

a) (2 points) List every choice variable and every source of income. Write the two period budget constraints and the intertemporal one, set up the Lagrangian, derive the first-order conditions and the Euler equation, and find c_1, c_2 and a.

Solution:

b) (1.5 points) The government cuts τ_1 to 0, finances the gap by selling bonds and repays them with interest by raising τ_2 to 30. Find the new c_1, c_2 and a, the change in private, public and national saving, and the assumptions the result needs.

Solution:

c) (1.5 points) Now she cannot borrow: $a \geq 0$. Solve her problem before and after the tax change of (b). Does Ricardian equivalence hold? What is her marginal propensity to consume out of the tax cut?

Solution:

Question 12. (1 point) Multiple choice. A household with $u(c) = c^{1-\sigma}/(1-\sigma)$, $\sigma > 0$, has income only in period 1 ($y_1 > 0$, $y_2 = 0$, no taxes, no initial wealth) and saves at rate r. When r rises, $\partial c_1/\partial r$:

A. is negative for every $\sigma > 0$: a higher r makes future consumption cheaper, so she saves more.

B. is positive iff $\sigma > 1$: for a saver, the income effect of r then outweighs the substitution effect.

C. is positive iff $\sigma < 1$: an elasticity of intertemporal substitution above one makes the income effect dominate.

D. is zero for every $\sigma > 0$: with $y_2 = 0$ there is no future income for a change in r to revalue.

E. is negative iff $\sigma > 1$: σ is the elasticity of intertemporal substitution, so a large σ makes substitution dominate.

Block 5 — Labour and leisure

(Aula 5 • Kurlat ch. 7 • 10 points)

Question 13. (4 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 7.3 "Consumption Taxes and Labor Supply", pp. 147–148 — statement reproduced verbatim, all items (a)–(e).*

Suppose a household solves the following problem:

$$\max_{c,l} \; u(c) + v(l)$$
$$s.t. \quad (1+\tau_c)\, c \leq w(1-\tau)(1-l)$$

where c is consumption, l is leisure and τ is an income tax. τ_c is a consumption tax, so that if the household wishes to consume c it must pay $(1+\tau_c)\, c$.

(a) Draw the household's budget constraint. What is the slope?

(b) Find the first-order conditions for the household's consumption-leisure decision. How do the effects of τ and τ_c relate to each other? Explain.

(c) If the household chooses c^* and l^*, how much tax revenue does the government collect?

(d) Suppose the government had originally set $\tau_c = 0$ and $\tau > 0$ and now wants to enact a tax reform that uses consumption taxes instead of income taxes. What would be the level of τ_c that leaves the decisions of the household unchanged?

(e) How much revenue does the government collect under the new system? How does it compare to the old system? Explain.

Added to the book's items (not in the book): in (d), also give the consumption tax that replaces a 30% income tax.

Solution:

Question 14. (5 points) Discursive. *Labour supply × taxes and transfers × Frisch elasticity.* A Brazilian household has one unit of time, splits it between leisure l and work $h = 1 - l$, and has utility $\ln c + \theta \ln l$, $\theta = 1.5$. It earns wage w, pays a labour-income tax τ and receives a lump-sum transfer T (Bolsa Família) that does not depend on its earnings. Let $\tilde{w} \equiv w(1-\tau)$. Baseline: $\tilde{w} = 0.7$, $T = 0.1$.

a) (2 points) Name the choice variables and the income sources, set up the Lagrangian with the time constraint $l \leq 1$, and derive the first-order conditions. Find labour supply $h(\tilde{w}, T)$ when it is interior, the condition for participation, and the baseline values of l, h and c.

Solution:

b) (1.5 points) Sign $\partial h / \partial \tilde{w}$ and explain it with income and substitution effects, separating $T = 0$ from $T > 0$. Then compute the change in h when (i) T rises to 0.2; (ii) with $w = 1$, τ rises from 0.3 to 0.4 and T stays at 0.1.

Solution:

c) (1.5 points) Compute the Frisch elasticity of labour supply (consumption held constant) and the uncompensated (Marshallian) elasticity with T fixed, at the baseline. Which one governs the response of hours to a temporary rise in wages, and which to a permanent one?

Solution:

Question 15. (1 point) Multiple choice. Kurlat's one-period search model: a share $U = 0.10$ of the labour force starts unemployed; matches are $m(V,U) = V^{1/2}U^{1/2}$; a vacancy costs χ and a filled job gives the firm $(1-\mu)(y-b)$; firms post vacancies until $q(V,U)(1-\mu)(y-b) = \chi$, with $q = m/V$; after hiring, unemployment is $U' = U - m(V,U)$. Initially $(1-\mu)(y-b)/\chi = 0.4$. Higher unemployment insurance b lowers this ratio to 0.3. Then:

A. V rises from 0.016 to 0.025 and U' falls from 0.06 to 0.05: a richer outside option makes workers easier to hire.

B. V falls from 0.016 to 0.009 and U' falls from 0.06 to 0.05: fewer vacancies are filled faster, so more workers find jobs.

C. V falls from 0.016 to 0.012 and U' falls from 0.06 to 0.05: vacancies fall in proportion to the firm's surplus.

D. V stays at 0.016 and U' rises from 0.06 to 0.07: benefits raise the wage but leave the vacancy decision unchanged.

E. V falls from 0.016 to 0.009 and U' rises from 0.06 to 0.07: a movement along a Beveridge-type relation.

Block 6 — General equilibrium: two periods, the welfare theorem, infinite horizon

(Kurlat ch. 9 (Aula 6) • 25 points)

Question 16. (5 points) Objective items.

a) (3 points) In Kurlat's economy of §9.1–§9.2 (price-taking households and firms, a market for every good at every date), which statement is correct?

A. Because the competitive equilibrium is Pareto efficient, its steady state must maximize consumption per worker (the golden rule).

B. An equilibrium in which the poorest households consume very little cannot be Pareto efficient, whatever the market structure.

C. If a firm's output imposes a cost on households for which no market exists, the equilibrium can be inefficient with price takers.

D. An unequal equilibrium with capital below the golden rule proves that markets are incomplete, so a planner would save more.

b) (2 points, T or F) In a two-period endowment economy with a representative household, $u = \ln c$ and endowments (y_1, y_2), news that y_2 will be higher raises the equilibrium real interest rate, while consumption in period 1 stays at y_1.

Question 17. (8 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 9.11 "Patience and Investment", p. 185 — statement reproduced verbatim, all items (a)–(c); item (d) is added by this list.*

Context from the book (added by this list, from Kurlat §9.3, pp. 172–177): *Infinite-period economy.* The representative household maximizes $\sum_{t=0}^{\infty} \beta^t [u(c_t)+v(l_t)]$ (9.3.1) subject to a budget constraint in every period; competitive firms pay factors their marginal products; goods market $F(K_t, L_t) = c_t + I_t$ (9.3.4); capital $K_{t+1} = K_t(1-\delta) + I_t$ (9.3.5). With labor supply fixed at $L_t = 1$ and $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, the equilibrium reduces to (pp. 174–175)

$$\frac{c_{t+1}}{c_t} = \left[\beta\left(1 + F_K(K_{t+1}, 1) - \delta\right)\right]^{\frac{1}{\sigma}}, \tag{9.3.14}$$

$$K_{t+1} = (1-\delta)\,K_t + F(K_t, 1) - c_t, \tag{9.3.15}$$

where c_t is consumption, K_t capital, F_K the marginal product of capital, β the discount factor, δ the depreciation rate and σ the curvature of utility. Consumption is constant, $\frac{c_{t+1}}{c_t} = 1$, only where (p. 175)

$$\left[\beta\left(1 + F_K(K_{t+1}, 1) - \delta\right)\right]^{\frac{1}{\sigma}} = 1 \tag{9.3.16}$$

$$\Rightarrow \; F_K(K_{t+1}, 1) - \delta = \frac{1}{\beta} - 1, \tag{9.3.17}$$

the vertical line of the phase diagram; the steady state K_{ss} satisfies (9.3.17). For comparison (p. 177, recalling the Solow model's (4.3.4)), the Golden Rule capital K_{gr}, which maximizes steady-state consumption, satisfies $F_K(K_{gr}, 1) = \delta$.

Suppose the production function is $F(K, L) = K^{\alpha} L^{1-\alpha}$, and the labor supply is exogenous and equal to $L = 1$. The representative household's preferences are given by:

$$\sum_{t=0}^{\infty} \beta^t \, \frac{c_t^{1-\sigma}}{1-\sigma}$$

and the depreciation rate is δ.

(a) Use equation (9.3.17) to find an expression for the level of capital that this economy will have in the steady state.

(b) Find an expression for the level of GDP that the economy will have in the steady state.

(c) Find an expression for the steady-state investment-to-GDP ratio. How does it depend on β? Explain.

Added parts (not in the book):

d) (3 of the 8 points) Compare the steady-state investment-to-GDP ratio of (c) with the golden-rule saving rate of the Solow model with the same technology and depreciation. Which is larger, and why does this not contradict the first welfare theorem? What happens as $\beta \to 1$? Evaluate everything at $\alpha = 0.4$, $\beta = 0.95$, $\delta = 0.08$ (the values of Exercise 9.12).

Solution:

Question 18. (12 points) Discursive. *A frontier economy: from one household's plan to the law of*

motion of capital. A stylized agricultural frontier (think of Brazil's Centro-Oeste) lasts two periods. A representative household has $\ln c_1 + \beta \ln c_2$, $0 < \beta < 1$, owns the initial capital $K_1 > 0$, has one unit of time in each period and gets no utility from leisure. It also owns the representative firm, which rents capital at R_t and labor at w_t and produces $Y_t = AK_t^{\alpha} L_t^{1-\alpha}$, $0 < \alpha < 1$. Capital depreciates fully after one period of use ($\delta = 1$). In period 1 the household can buy new capital K_2 (one unit of output each) and trade a one-period bond a_1 at the real rate r.

a) (3 points) Write both budget constraints, listing every income source, and the present-value constraint. Solve the firm's problem and find its profit. Derive the household's first-order conditions for *every* choice variable.

Solution:

b) (3 points) Define a competitive equilibrium (allocation, prices, conditions). Use market clearing to find K_2, c_1, the equilibrium saving rate K_2/Y_1 and r. Check c_1 against the household's present-value constraint.

Solution:

c) (3 points) Solve the planner's problem for this economy and compare with (b). Which theorem have you verified, and what plays the role of the interest rate in the planner's problem?

Solution:

d) (3 points) Now let the economy last forever: $\sum_t \beta^t \ln c_t$, same technology, $\delta = 1$, $L_t = 1$. Show

that the rule $K_{t+1} = \alpha\beta A K_t^{\alpha}$, $c_t = (1-\alpha\beta) A K_t^{\alpha}$ satisfies the Euler equation at every date and the transversality condition. Draw the map $K_{t+1}(K_t)$ with the 45-degree line, find the steady state and its stability, and relate the result to Solow ($\delta = 1$, $n = 0$) and to the golden rule.

Solution:

Block 7 — Money and inflation: aggregates, money demand, the quantity theory, seigniorage

(Kurlat chs. 10–11 (Aula 7) • 25 points)

Question 19. (5 points) Objective items.

a) (3 points) A Brazilian saver holds a fixed-rate (*prefixado*) bond. Income tax takes $\tau = 15\%$ of the *nominal* interest. The real rate is $r = 7\%$ and the Fisher effect holds ($i = r + \pi$). Anticipated inflation rises from 5% to 10%. The after-tax real return $(1-\tau)i - \pi$ goes:

A. from 5.95% to 5.95%

B. from 5.2% to 4.45%

C. from 7% to 7%

D. from 12% to 17%

b) (2 points, T or F) With Baumol–Tobin money demand and flexible prices, a permanent, anticipated rise in the growth rate of money raises inflation one for one in the new steady state and leaves real money balances unchanged.

Question 20. (8 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 11.6 "Seignorage with High Inflation", pp. 219–221 — reproduced verbatim; this list asks items (a)–(j) only (the book's items (k)–(u) are omitted).*

Context from the book (added by this list, from Kurlat §11.1–§11.3, pp. 207–216): *Interest rates.* A lender gives one dollar in period t and gets back $1 + i_{t+1}$ dollars in $t + 1$: i_{t+1} is the nominal interest rate between t and $t + 1$ (p. 207). With price level P_t (written p_t in the money-market condition) and inflation π_{t+1}, $1 + \pi_{t+1} = \frac{P_{t+1}}{P_t}$, the real interest rate r_{t+1} satisfies (p. 208)

$$1 + r_{t+1} = \frac{1 + i_{t+1}}{1 + \pi_{t+1}} \quad \Rightarrow \quad r_{t+1} \approx i_{t+1} - \pi_{t+1}, \tag{11.1.1}$$

the Fisher equation (the approximation is accurate when π_{t+1} is small); the nominal rate is the opportunity cost of holding money. *Money market* (p. 209): equilibrium requires money supply to equal money demand,

$$M^S = m^D(Y, i) \cdot p, \tag{11.2.1}$$

where $m^D(Y, i)$ is real money demand, Y real GDP and p the price level; the central bank chooses M^S by choosing the monetary base M^B and understanding the money multiplier. *Seignorage* (§11.3, p. 216): in the government budget constraint (11.3.1), the increase in the monetary base $M^B_{t+1} - M^B_t$ is a source of revenue, like debt that pays no interest and is never repaid.

Suppose that money demand is given by:

$$m^D(Y_t, i_{t+1}) = Y_t \cdot (1 + i_{t+1})^{-\mu}$$

where μ is a parameter. This is sometimes known as the Cagan money-demand function. Equilibrium in the money market is given by:

$$M^S_t = Y_t \cdot (1 + i_{t+1})^{-\mu} p_t$$

The real economy is in a steady state so that:

$$Y_t = Y_{ss}$$
$$r_{t+1} = r_{ss}$$

The government obtains seignorage revenue S_t (in real terms) by expanding the monetary base. S_t is given by:

$$S_t = \frac{M^B_t - M^B_{t-1}}{P_t} \tag{11.4.2}$$

Let ω be the money multiplier, so that $M^S = \omega M^B$.
Suppose that the rate of growth of the money supply is constant at rate γ, i.e.:

$$M^S_t = (1 + \gamma)\, M^S_{t-1} \tag{11.4.3}$$

(a) What will be the rate of inflation?

(b) What will be nominal interest rate?

(c) Find an expression for the level of real money balances. How does it depend on γ? Why?

(d) Solve equation (11.4.3) for M^S_{t-1} and replace this in (11.4.2) to obtain an expression for seignorage revenue S_t in terms of γ and $\frac{M^S_t}{P_t}$. How does S_t depend on γ? How does it depend on $\frac{M^S_t}{P_t}$? Why?

(e) Replace the value of $\frac{M^S_t}{P_t}$ that you found in part (c) into the expression for S_t that you found in part (d) to obtain an expression for seignorage revenues S_t in terms of γ, Y_{ss}, r_{ss}, ω and μ.

(f) Assume the following parameter values:

$$Y_{ss} = 1$$
$$\mu = 4$$
$$r_{ss} = 0.0025$$
$$\omega = 5$$

Note that $\mu = 4$ is close to the value that has been empirically estimated in some high inflation countries when the time period is one month. To be consistent, r_{ss} should be interpreted as a monthly real interest rate and Y_{ss} as monthly GDP.

(g) Plot S_t against γ for values of γ between 0 and 0.8.

(h) What is the monthly rate of growth of the money supply that maximizes seignorage revenue? Call this γ^*.

(i) Why is seignorage revenue decreasing in γ for $\gamma > \gamma^*$?

(j) What monthly inflation rate does $\gamma = \gamma^*$ imply?

Added to the book's items (not in the book): draw the plot of part (g). Careful with notation: here μ is the Cagan semi-elasticity, not money growth (γ).

Solution:

Question 21. (12 points) Discursive. *From the monetary base to the price level: a credible disinflation.* The Banco Central do Brasil (BCB) controls the base B. The public holds currency and deposits in the ratio $c = C/D = 0.25$; banks hold reserves (required plus excess) in the ratio $\theta = R/D = 0.20$. Real money demand follows Baumol–Tobin: spending Y per period is withdrawn in N equal trips to the bank, each costing F in real terms. Prices are flexible, real output grows at $g = 2\%$, the real rate is constant at $r = 4\%$ and M1 grows at $\mu = 6\%$ (illustrative values).

a) (3 points) Derive the money multiplier $m = M1/B$ and compute it. The BCB cuts reserve requirements so that θ falls to 0.15: by how much does M1 change for a given base? Why is m not a constant the BCB can rely on?

Solution:

b) (3 points) Derive Baumol–Tobin money demand, its income and interest elasticities, and the implied velocity.

Solution:

c) (3 points) In a steady state with constant i, show that $\pi = \mu - \eta g$, where η is the income elasticity of money demand. Compute π and i (exact Fisher) now, and the money growth the BCB needs for an inflation target of 3%.

Solution:

d) (3 points) At date T the BCB announces, credibly, that M1 will grow at 4% from now on; M_T itself does not change. What happens to the price level at T and afterwards? Draw $\ln P_t$. Is money superneutral here? What one-off operation would avoid any jump in P?

Solution:

Block 8 — The New Keynesian AS–AD model: microfoundations

(Benigno (2015) §§1–5 (Aula 8) • 25 points)

Notation for Blocks 8–9 (Benigno 2015). Lower-case letters are log deviations from an efficient steady state; bars denote the long run. Households: $u(C) - v(L) + \beta\{u(\bar{C}) - v(\bar{L})\}$, $u_c = C^{-1/\tilde{\sigma}}$, $v = L^{1+\eta}/(1+\eta)$, $\rho \equiv -\ln\beta$; resources $y = s_c c + g$, $\sigma \equiv \tilde{\sigma} s_c$. Firms: technology $Y(j) = AL(j)$, desired markup μ (monopoly power and taxes, $\mu \simeq \mu_\theta + \tau + \tau_c$, with τ collecting labor, payroll and sales taxes); a fraction α of prices is preset at p^e. AD, Benigno's eq. (21), and AS,

eq. (20), with $\kappa = \frac{(1-\alpha)(\sigma^{-1}+\eta)}{\alpha}$:

$$y = \bar{y}_n + (g - \bar{g}) - \sigma[i - (\bar{p} - p) - (\bar{\tau}_c - \tau_c) - \rho], \tag{AD}$$

$$p - p^e = \kappa(y - y_n). \tag{AS}$$

Calibration when numbers are asked (Benigno, p. 515): $\alpha = 0.66$, $\sigma = 0.5$, $\eta = 0.2$, so $\sigma^{-1} + \eta = 2.2$, $\kappa = 1.133$.

Question 22. (5 points) Objective items.

a) (3 points) Why does Benigno's AD curve slope downward in the (y, p) plane?

A. A higher p lowers real balances M/P, so the interest rate that clears the money market rises and spending falls.

B. A higher p, given $\bar{p}$, raises expected inflation $\bar{p} - p$, which lowers the real rate and pushes households to consume.

C. A higher p lowers real balances, so the central bank's money-supply rule raises i and households save more today.

D. A higher p, given i and $\bar{p}$, lowers expected inflation $\bar{p} - p$, so the real rate rises and households save more.

b) (2 points, T or F) News that the long-run desired markup $\bar{\mu}$ will be higher (current μ unchanged) shifts the AS curve up today, because firms that reset prices raise them in anticipation.

Question 23. (8 points) *Source: built on Benigno, P. (2015), "New-Keynesian Economics: An AS–AD View", §4.1–§4.4, pp. 507–509 — the article has no exercises; the statement below is ours.* Natural versus efficient output: Benigno's eqs. (14)–(15) and (19), the topic of Lista 7 Q3. Set $g = 0$.

Context from the article (added by this list, from Benigno 2015, §3–§4.4, pp. 506–509): *Firms.* Firm j faces demand $Y(j) = \left(\frac{P(j)}{P}\right)^{-\theta}(C+G)$ (9), with $\theta > 0$ the elasticity of substitution among goods; its only input is labor, $Y(j) = AL(j)$, with A productivity; its profits are $\Pi(j) = (1-\tau_y)P(j)Y(j) - (1+\tau_w)WL(j)$ (10), with τ_y the tax on sales and τ_w the tax on labor costs. A fraction α of firms keeps its price at P^e; the others set a markup over marginal cost, $P(j) = (1+\tilde{\mu})\frac{W}{A}$ (11), with $\tilde{\mu} \equiv \frac{\theta}{\theta-1}\frac{1+\tau_w}{1-\tau_y} - 1$. *Households* supply labor where $\frac{v_l(L)}{u_c(C)} = \frac{1-\tau_l}{1+\tau_c}\frac{W}{P}$ (7), with τ_c the consumption tax and τ_l the labor-income tax, $u_c(C) = C^{-\tilde{\sigma}^{-1}}$ and $v(L) = L^{1+\eta}/(1+\eta)$. The aggregate markup combines monopoly power and taxes:

$$\mu \equiv \mu_\theta \frac{(1+\tau_w)(1+\tau_c)}{(1-\tau_y)(1-\tau_l)} - 1, \qquad \mu_\theta \equiv \frac{\theta}{\theta-1}. \tag{13}$$

Natural output Y_n is output when all firms can adjust prices; with $Y = C + G$ and $Y = AL$ it solves

$$\frac{(Y_n/A)^\eta}{(Y_n - G)^{-\tilde{\sigma}^{-1}}} = \frac{A}{1+\mu}. \tag{14}$$

Efficient output Y_e maximizes $u(C) - v(L)$ subject to $Y = C + G$ and $Y = AL$; its optimality condition is $\frac{u_c(Y_e - G)}{v_l(Y_e/A)} = \frac{1}{A}$. Log-linearize with $\ln(1+\mu) \simeq \mu$ and $\sigma \equiv \tilde{\sigma} s_c$, where $s_c = C/Y$. The article's log-linear results, (15) for y_n and (19) for y_e, are what item (a) asks you to derive, so they are not restated here.

What the article states (p. 509): "A comparison of (19) with (15) shows that productivity and public spending shift the two definitions of output in equal proportion. Mark-up shocks, however, are a source of inefficiencies, as they shift the natural but not the efficient level of output."

a) (3 points) Derive natural output y_n (flexible prices) and efficient output y_e (planner), and the gap between them.

Solution:

b) (3 points) Draw the labor market in the plane (hours l, real wage $w - p$) with $a = 0$, marking natural and efficient hours, and shade the welfare loss. Compute $y_n - y_e$ and the loss for $\mu = 5\%$ at the calibration.

Solution:

c) (2 points) Which of α, θ, a, g and the tax rates move $y_n - y_e$? Can monetary policy close it? What could?

Solution:

Question 24. (12 points) Discursive. *Brazil's consumption-tax reform in the AS–AD model.* The reform replaces several taxes with a value-added tax that phases in over years. Treat each fiscal change below as a deviation from an efficient steady state with $p^e = \bar{p} = 0$, $g = \bar{g} = 0$ and $i = \rho$.

a) (4 points) The household's budget is $(1 + \tau_c)PC + \frac{(1+\bar{\tau}_c)\bar{P}\bar{C}}{1+i} = WL + \frac{\bar{W}\bar{L}}{1+i} + T$. Derive the Euler equation, log-linearize it, and use $y = s_c c + g$ and long-run price flexibility to obtain the AD (21). Give its slope and explain it.

Solution:

b) (4 points) For each change, say which curve(s) move, in which direction, and by how much horizontally: (i) a temporary rise in payroll tax, $d\tau > 0$; (ii) an announced long-run rise, $d\bar{\tau} > 0$; (iii) an announced long-run rise in the consumption tax, $d\bar{\tau}_c > 0$; (iv) a temporary rise in the consumption tax, $d\tau_c > 0$.

Solution:

c) (4 points) Solve case (ii) with i, $\bar{p}$ and p^e fixed: find dy, dp, and both gaps $y - y_n$, $y - y_e$. Draw the diagram and evaluate for $d\bar{\tau} = 1$ percentage point at the calibration. Which way should i move to restore $p = p^e$, and is there a trade-off?

Solution:

Block 9 — AS–AD policy: shocks, divine coincidence, optimal policy, multipliers, ZLB

(Benigno (2015) §§6–12 (Aula 9) • 25 points)

Question 25. (5 points) Objective items.

a) (3 points) Which shock leaves the central bank a trade-off between $p = p^e$ and $y = y_e$, and why?

A. A temporary rise in the current markup μ: it lowers y_n but not y_e, so $p = p^e$ and $y = y_e$ cannot both hold on the AS.

B. A temporary rise in productivity a: it lowers p along the AD, so a cut in i cannot bring p back to p^e without a gap.

C. News of a higher long-run markup $\bar{\mu}$: it shifts the AS up today and separates y_n from y_e in the current period.

D. News of higher long-run productivity $\bar{a}$: it lifts demand above y_e, and raising i lowers p but leaves a positive gap.

b) (2 points, T or F) A temporary rise in the labor-income tax τ (with i, $\bar{p}$, p^e fixed) lowers output and also lowers the output gap $y - y_n$.

Question 26. (8 points) *Source: built on Benigno, P. (2015), "New-Keynesian Economics: An AS–AD View", §8, pp. 514–516 — the article has no exercises; the statement below is ours.* The spending multiplier: Benigno's eqs. (22)–(23), Tables 1–2. Take $i = \rho$, $\bar{p} = p^e = 0$, $a = \mu = 0$, and a temporary rise in g with $\bar{g}$ fixed.

Context from the article (added by this list, from Benigno 2015, §8, pp. 514–516): Keeping only the terms of AD (21) that fiscal policy affects, $y = g + \bar{y}_n - \bar{g} - \sigma[p - (\bar{\tau}_c - \tau_c)]$, substituting AS (20) for p, and using $\mu \approx \mu_\theta + \tau_l + \tau_w + \tau_y + \tau_c$ (and the same with bars) in the natural level of output, Benigno obtains

$$y = m_g g - m_{\bar{g}} \bar{g} - m_\tau \tau - m_{\bar{\tau}} \bar{\tau} - m_{\tau_c} \tau_c + m_{\bar{\tau}_c} \bar{\tau}_c, \tag{22}$$

where $\tau \equiv \tau_l + \tau_w + \tau_y$ collects every tax except the consumption tax τ_c (likewise $\bar{\tau}$), and each m is a positive coefficient that depends on σ, η and κ. In the article, Table 1 (p. 515) gives these coefficients in closed form, eq. (23) (p. 515) gives the output gap $y - y_n$ obtained by subtracting (15) from (22), and Table 2 (p. 516) evaluates them for eight calibrations (α, σ, η). Items (a)–(c) ask you to derive and evaluate exactly these, so they are not restated here.

What the article states (p. 514): "In particular, an increase in current public expenditure, $g \uparrow$, increases output. The increase is one-to-one in the special case of linear disutility of labor, i.e. $\eta = 0$; otherwise the increase is less than proportional."

a) (3 points) Derive the output multiplier $m_g = dy/dg$ from the AD and AS curves. Show that $m_g \leq 1$, with equality iff $\eta = 0$.

Solution:

b) (3 points) Find the multiplier on the gap $y - y_n$ and on the efficient gap $y - y_e$. Why are they so much smaller than m_g?

Solution:

c) (2 points) Evaluate κ, m_g, the gap multiplier and $1 - m_g$ at the calibration, and compare with Table 2.

Solution:

Question 27. (12 points) Discursive. *The Copom's three meetings.* Start from the efficient steady state: $y_n = y_e = 0$, $p^e = \bar{p} = 0$, $i = \rho = 2\%$. The welfare loss is $L = \frac{1}{2}(y - y_e)^2 + \frac{\theta}{2\kappa}(p - p^e)^2$ (eq. 33) with $\theta = 8$; use the calibration of Block 8 and work in percentage points.

a) (4 points) Meeting 1: news that long-run productivity will rise, $d\bar{a} = 1$, current a unchanged (say, gains from agricultural technology expected to last). Which curves move? Find y and p if the bank keeps $i = \rho$, and the optimal i. Is there a trade-off? Contrast with a temporary rise in a.

Solution:

b) (5 points) Meeting 2: a temporary markup shock $\mu = 5$ (say, a fuel-tax rise), $\bar{\mu} = 0$. Find the outcome at $i = \rho$. Then minimize L subject to the AS and derive the targeting rule, y^*, p^*, and the rate i^*, comparing it with r_n and ρ. Draw.

Solution:

c) (3 points) Meeting 3: news that long-run productivity will fall, $d\bar{a} = -5$, with $\mu = a = 0$. Show that the required rate is negative. With $i = 0$, find the gap and p. What can the bank still do, and on what does it depend?

Solution: